# BPL_TEST2_Batch - demo

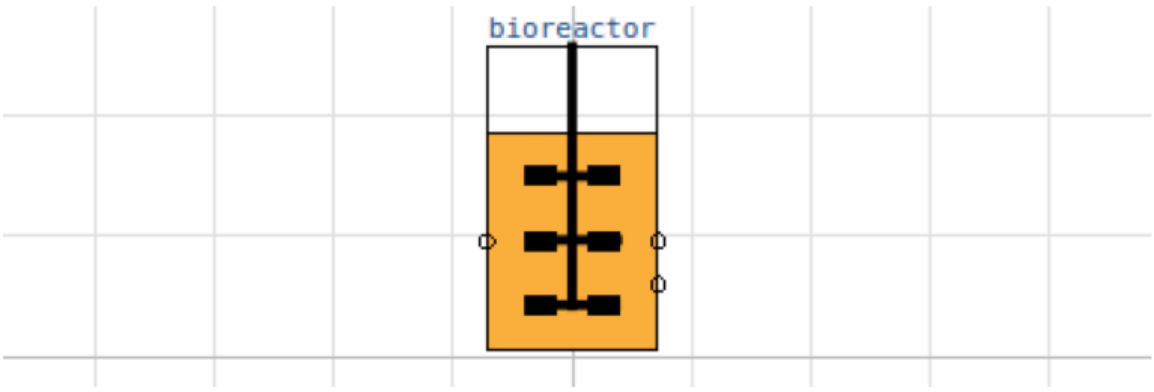
In [19]:
```
run -i BPL_TEST2_Batch_fmpy_explore.py
```

```
Windows - run FMU pre-compiled JModelica 2.14

Model for the process has been setup. Key commands:
 - par()       - change of parameters and initial values
 - init()      - change initial values only
 - simu()      - simulate and plot
 - newplot()   - make a new plot
 - show()      - show plot from previous simulation
 - disp()      - display parameters and initial values from the last simulation
 - describe()  - describe culture, broth, parameters, variables with values/units

Note that both disp() and describe() takes values from the last simulation
and the command process_diagram() brings up the main configuration

Brief information about a command by help(), eg help(simu)
Key system information is listed with the command system_info()
<Figure size 984.252x787.402 with 0 Axes>
```

In [20]:
```
%matplotlib inline
plt.rcParams['figure.figsize'] = [25/2.54, 20/2.54]
```

In [21]:
```
process_diagram()
```

```
No processDiagram.png file in the FMU, but try the file on disk.
```



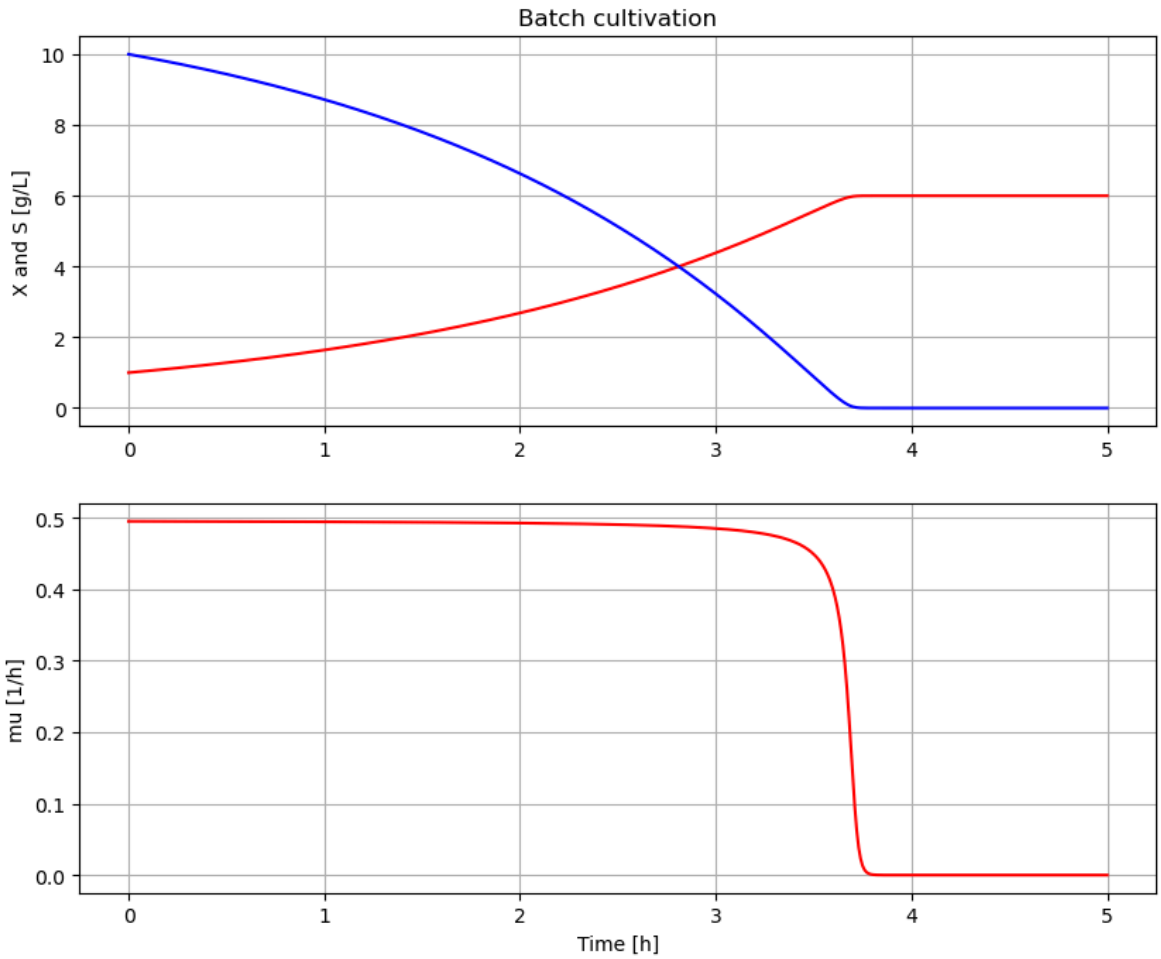

In [22]:
```
describe('process')
```

```
None
```

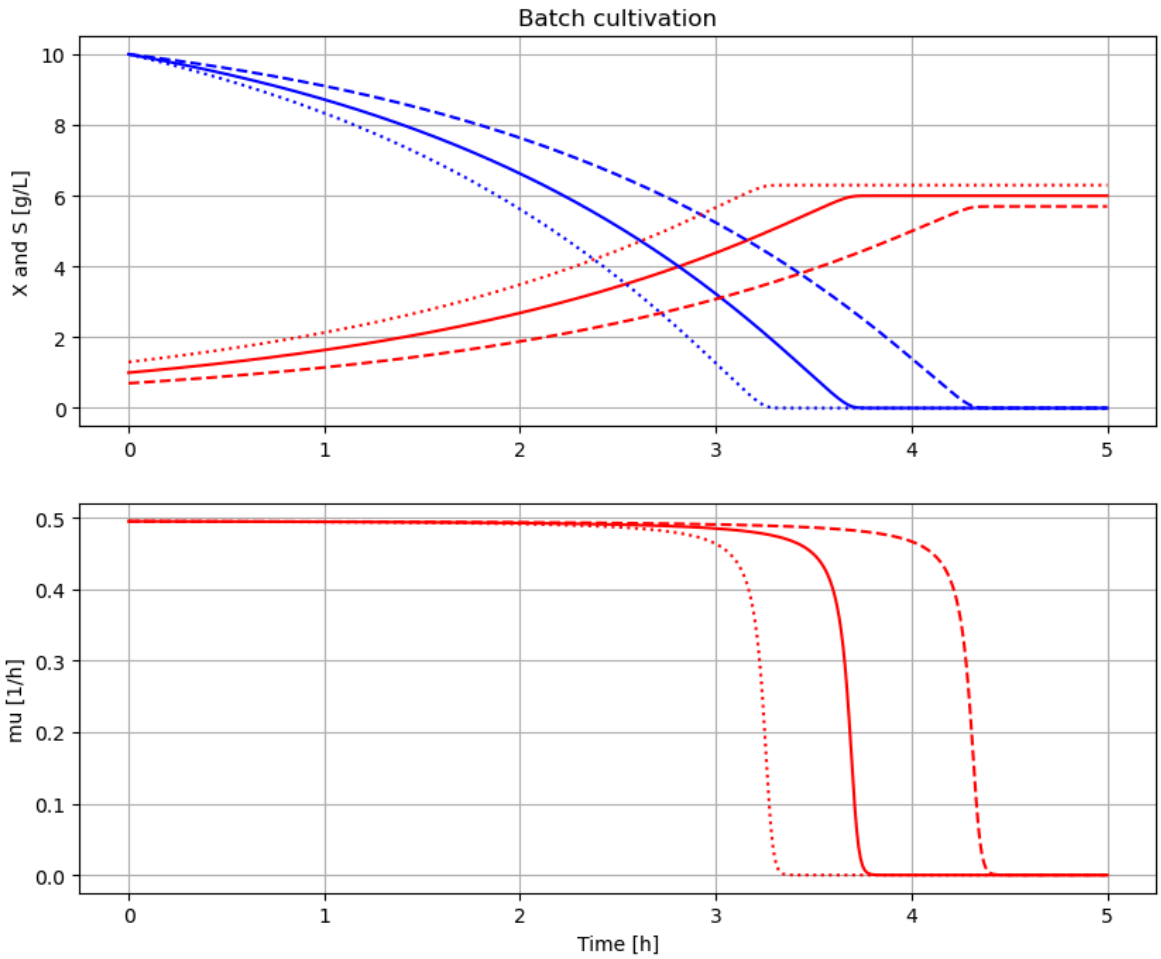
In [23]:
```
describe('culture')
```

```
Simplified text book model - only substrate S and cell concentration X
```

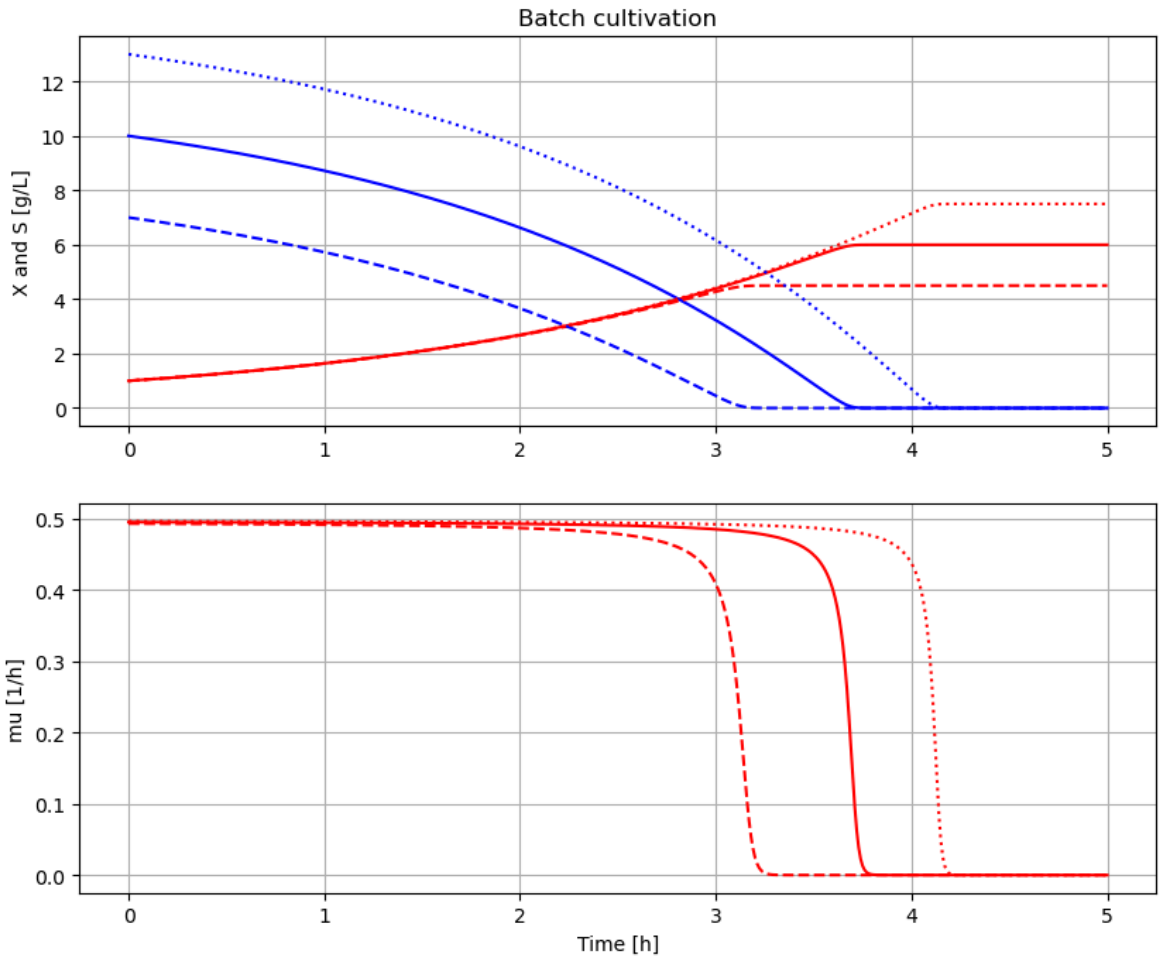
In [24]:
```
# Simulation with default values of the process
newplot(plotType='TimeSeries')
simu()
```



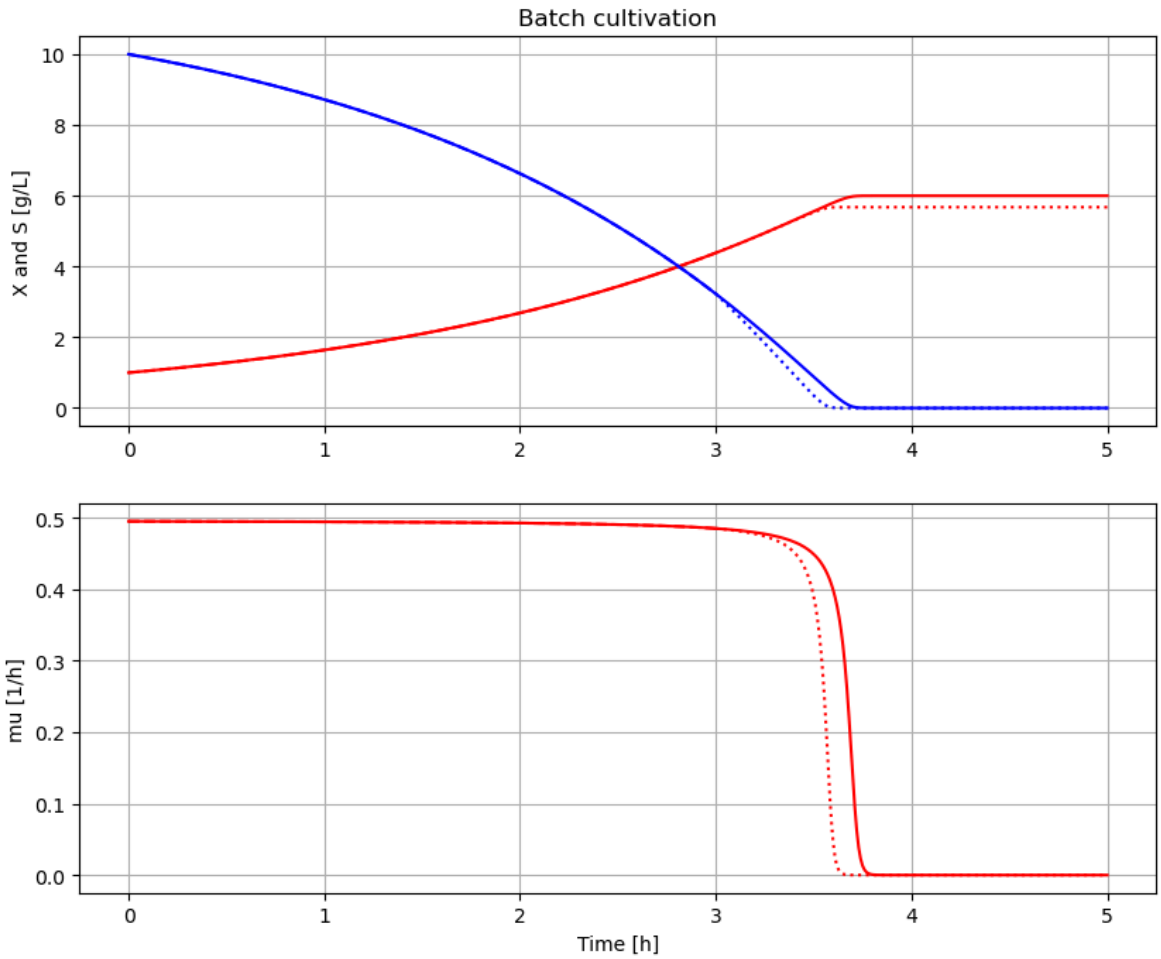

In [25]:
```
# Simulation were initial value of biomass VX_start is varied
newplot(plotType='TimeSeries')
for value in [1.0, 0.7, 1.3]: init(VX_start=value); simu(5)

# Restore default value of VX_start
init(VX_start=1.0)
```




In [26]:
```
# Simulation were initial value of substrate VS_0 is varied
newplot(plotType='TimeSeries')
for value in [10, 7, 13]: init(VS_start=value); simu(5)

# Restore default value of VS_0
init(VS_start=10)
```




In [27]:
```
# Simulation where metabolism is changed after 3 hours
newplot(plotType='TimeSeries')
simu(5)

simu(3)
par(Y=0.4, qSmax=1.0/(0.4/0.5)); simu(2, 'cont')

# Restore default value of Y and qSmax
par(Y=0.5, qSmax=1.0)
```




In [28]:
```
disp('culture')
```

```
Y : 0.5
qSmax : 1.0
Ks : 0.1
```

In [29]:
```
describe('mu')
```

```
Cell specific growth rate variable : 0.0 [ 1/h ]
```

In [30]:
```
describe('parts')
```

```
['bioreactor', 'bioreactor.culture', 'MSL']
```

In [31]:
```
describe('MSL')
```

```
MSL: none
```

In [32]:
```
system_info()
```

```
System information
 -OS: Windows
 -Python: 3.12.11
 -Scipy: not installed in the notebook
 -FMPy: 0.3.27
 -FMU by: JModelica.org
 -FMI: 2.0
 -Type: CS
 -Name: BPL.Examples_TEST2.Batch
 -Generated: 2026-04-09T15:38:29
 -MSL: 3.2.2 build 3
 -Description: Bioprocess Library version 2.3.2
 -Interaction: FMU-explore for FMPy version 1.0.3
```

In [33]:
```
#!lsb_release -a
```

In [34]:
```
SDG('explain')
```

```
"Soli Deo Gloria"
 It is latin and means "To the honour of God".
 The great composer Johan Sebastian Bach used to end his compositions with this s
mall remark SDG.
 And I like to do that too :).
```

In [ ]: